UNIVERSIDAD DE LAS FUERZAS ARMADAS – ESPE
Department of Computer Science
Software Engineering
36862 – Object-Oriented Programming

UNIT 1 PROJECT · TEAM WORK

Product Backlog: ProvStock

Prov&Tec Management, a Comprehensive Sales and Inventory System

Team	ByteWorks
Members	Andrés Pachacama, Carlos Parra, Matias Paucar
NRC	36862
Instructor	Edison Lascano, PhD
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Version	1.3

Revision History

Table 1
Document Versions

Version	Date	Team	Author	Description of Changes
1.0	October 7, 2026	Team ByteWorks	Matias Paucar	Initial Product Backlog with the nine items RF01 to RF09, their tasks, owners, and priorities.
1.1	October 20, 2026	Team ByteWorks	Andrés Pachacama	Priorities aligned with the MoSCoW scale of the SRS v1.2; RF08 renamed to Stock Alerts; tasks aligned with the functional requirements of the SRS v1.2; references to the SRS requirements and to the use cases added.
1.2	October 22, 2026	Team ByteWorks	Carlos Parra	Status of every item updated to Done after the implementation of ProvStock in Java, including data persistence; items, tasks, and priorities unchanged.
1.3	October 26, 2026	Team ByteWorks	Andrés Pachacama	Aligned with the SRS v1.3: the reports of RF09 are saved as CSV files instead of PDF, and the data is stored in JSON files, one per class; items, owners, and priorities unchanged, and every item still Done.

1. Backlog Summary

Table 2 lists the nine Product Backlog items, their owner, their priority, and their status.

Table 2
Product Backlog Items

ID	Item	Owner	Priority	Status
RF01	Access Management	Carlos Parra	Must have	✓ **Done**
RF02	Customer Management	Carlos Parra	Must have	✓ **Done**
RF03	Product Management	Carlos Parra	Must have	✓ **Done**
RF04	Sales Management	Matias Paucar	Must have	✓ **Done**
RF05	Inventory Management	Matias Paucar	Must have	✓ **Done**
RF06	Supplier and Purchase Management	Matias Paucar	Must have	✓ **Done**
RF07	Sales History	Andrés Pachacama	Should have	✓ **Done**
RF08	Stock Alerts	Andrés Pachacama	Should have	✓ **Done**
RF09	Reports	Andrés Pachacama	Should have	✓ **Done**

2. Priority Scale

Priorities follow the MoSCoW scale used in the SRS v1.3:

- **Must have:** essential for the first release.
- **Should have:** important, but the system is usable without it.
- **Could have:** desirable if time allows.

3. Backlog Items

RF01. ACCESS MANAGEMENT	
Tasks	☑ Define three user roles (Salesperson, Warehouse Clerk, and Owner), each with its own permissions. ☑ Allow users to log in and log out with a username and password. ☑ Allow the owner to create, update, and deactivate user accounts and to assign a role to each one. ☑ Allow each user to change their own password. ☑ Allow the owner to reset the password of a user who forgot it.
SRS reference	FR-01.1 – FR-01.7
Use cases	UC-01 Log in, UC-02 Log out, UC-03 Change password, UC-04 Manage user accounts
Owner	Carlos Parra
Priority	Must have
Status	**✓ Done — All tasks completed.**

RF02. CUSTOMER MANAGEMENT	
Tasks	☑ Register new customers, validating that the identification number is unique and has a valid length. ☑ Update the data of existing customers. ☑ Search for customers by identification number or name and view their information. ☑ Provide a default “Final Consumer” customer for sales without personal data. ☑ Allow the owner to deactivate customers.
SRS reference	FR-02.1 – FR-02.6
Use cases	UC-05 Manage customers
Owner	Carlos Parra
Priority	Must have
Status	**✓ Done — All tasks completed.**

RF03. PRODUCT MANAGEMENT	
Tasks	☑ Register products with their minimum stock level and their location in the warehouse. ☑ Update product information, except the current stock. ☑ Search for products by code, name, or category and show their price, stock, and location. ☑ Deactivate products that are no longer sold.
SRS reference	FR-03.1 – FR-03.5
Use cases	UC-06 Search products, UC-07 Manage products
Owner	Carlos Parra
Priority	Must have
Status	**✓ Done — All tasks completed.**

RF04. SALES MANAGEMENT	
Tasks	☑ Register sales with their customer, products, and quantities; record the date, time, salesperson, and a consecutive number automatically. ☑ Prevent selling more than the available stock and calculate the subtotal, VAT, and total of each sale. ☑ Allow the owner to void a sale registered by mistake, entering a reason. ☑ Allow the owner to update the VAT rate.
SRS reference	FR-04.1 – FR-04.9
Use cases	UC-08 Register sale, UC-09 Void sale, UC-10 Update VAT rate
Owner	Matias Paucar
Priority	Must have (void sale and VAT rate update: Should have)
Status	**✓ Done — All tasks completed.**

RF05. INVENTORY MANAGEMENT	
Tasks	☑ Update the stock automatically when a sale, a purchase, or a voided sale is registered. ☑ Allow the warehouse clerk and the owner to adjust the stock manually with a reason, including the load of the initial inventory. ☑ Show the current stock of all products and highlight those at or below the minimum level. ☑ Record every stock movement with its date, type, quantity, and user.
SRS reference	FR-05.1 – FR-05.6
Use cases	UC-11 Update stock, UC-12 View inventory, UC-13 Adjust stock
Owner	Matias Paucar

Priority	Must have
Status	✓ Done — All tasks completed.

RF06. SUPPLIER AND PURCHASE MANAGEMENT	
Tasks	☑ Allow the owner to register, update, and deactivate suppliers, validating that the RUC is unique. ☑ Search for suppliers by RUC or name. ☑ Register purchases with their supplier, products, quantities, and unit costs; calculate a consecutive number and the total cost automatically.
SRS reference	FR-06.1 – FR-06.5
Use cases	UC-14 Manage suppliers, UC-15 Register purchase
Owner	Matias Paucar
Priority	Must have
Status	**✓ Done — All tasks completed.**

RF07. SALES HISTORY	
Tasks	☑ Allow the owner to view all registered sales, including voided ones, with their date, customer, salesperson, total, and status. ☑ Allow the owner to filter the sales by date range, salesperson, customer, or status. ☑ Show the details of a selected sale.
SRS reference	FR-07.1 – FR-07.3
Use cases	UC-16 View sales history
Owner	Andrés Pachacama
Priority	Should have
Status	**✓ Done — All tasks completed.**

RF08. STOCK ALERTS	
Tasks	☑ Check whether a product is at or below its minimum stock level after every stock change and every time a user logs in. ☑ Show the alerts to the warehouse clerk and the owner, with the number of active alerts in the main menu. ☑ Remove the alert automatically when the stock rises above the minimum level.
SRS reference	FR-08.1 – FR-08.4
Use cases	UC-17 Check minimum stock, UC-18 Receive stock alerts
Owner	Andrés Pachacama

Priority	Should have
Status	✓ Done — All tasks completed.

RF09. REPORTS	
Tasks	☑ Generate a sales report for a selected period. ☑ Generate an inventory report with the value of the stock. ☑ Generate a purchase report per supplier for a selected period. ☑ Generate a report of slow-moving products (products without sales in a selected period). ☑ Save every generated report as a CSV file that can be opened with Excel.
SRS reference	FR-09.1 – FR-09.5
Use cases	UC-19 Generate reports, UC-20 Export report to CSV
Owner	Andrés Pachacama
Priority	Should have (slow-moving products report and export to CSV: Could have)
Status	✓ Done — All tasks completed.

4. Progress by Team Member

Table 3
Progress by Team Member

Team member	Items	Done
Carlos Parra	RF01, RF02, RF03	**3 / 3**
Matias Paucar	RF04, RF05, RF06	**3 / 3**
Andrés Pachacama	RF07, RF08, RF09	**3 / 3**
Total	**RF01 – RF09**	**9 / 9**